

## DESIGN OF COASTAL STRUCTURES

## COASTAL STRUCTURES

- **PORT STRUCTURES**
  - \*Breakwaters
  - \*Quays(wharves), piers(jetties), quaywalls (bulkheads)
  - \* *Transit sheds, warehouses, railways and roads*
- **COASTAL DEFENSE STRUCTURES**
  - \*Seawalls, revetments
  - \*Groins
- **OTHER MARINE STRUCTURES**
  - \*pipelines
  - \*offshore platforms
  - \*sea outfall structures

## Design of Breakwaters

- **Lay-out of the breakwater**
  - Location of entrance
  - The water area to be covered by breakwaters
  - The tranquillity of harbour against waves
- Minimum length economy side of view  
At the same time it should provide sufficient water area in the harbour
- Basically this may be achieved by laying out the breakwater perpendicular to the predominant wave direction.
- Usually they extend perpendicularly from shore to a sufficient depth at the offshore and then cross the predominant wave direction by a change of alignment
- Cross-section of the breakwater

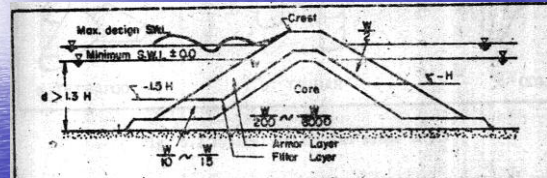
## layout



## Breakwaters

- Sloping type (rubble mound breakwaters)
- Vertical wall (upright)
- Floating
- Piled

## Sloping type



## vertical

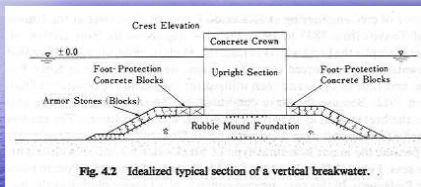
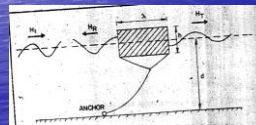
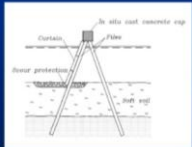


Fig. 4.2 Idealized typical section of a vertical breakwater.

## Floating breakwaters



## Piled breakwater



Piled breakwaters consist of an inclined or curtain wall mounted on pile work.

The type is applicable in less severe wave climates on site with weak and soft subsoils with very thick layers.



## Design of Rubble Mound Breakwaters

Rubble-mound breakwaters are the most common type of breakwater in Turkey.

In its most simple shape it is a mound of stones.

Armour layer



Permeable  
expensive

filter layer+ core layer

## Design of Rubble Mound Breakwaters

Concrete armour layer

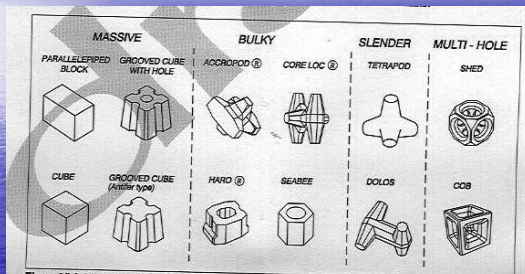


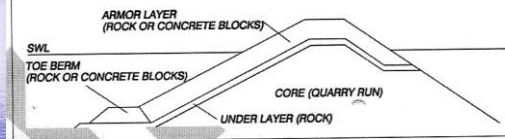
Figure VI-2-24. Examples of concrete armor units

## Concrete armor units



## Design of Rubble Mound Breakwaters

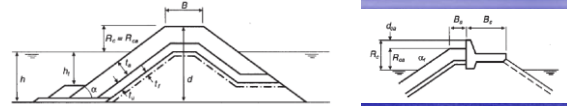
### CONVENTIONAL MULTI-LAYER RUBBLE-MOUND BREAKWATER



Elements of rock structures for which the structural response under waves should be analysed include:

- armour layer at seaward side, crest, rear-side and breakwater head
- front side toe stability and (need for) scour protection
- filter layers, core material and geotextiles
- crown wall.
- Crest height
- Crest width

## Design of Rubble Mound Breakwaters



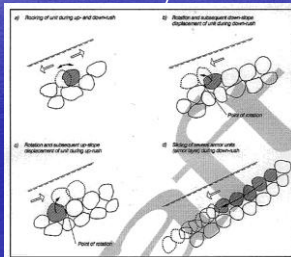
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## Design of Rubble Mound Breakwaters

Wave forces acting on a rubble-mound slope can cause armor unit movement. This is called as hydraulic instability.

Armor unit movements:



## Design of Rubble Mound Breakwaters

The complicated flow of waves impacting armor layers makes it impossible to calculate the flow forces acting on armor units.

Moreover, the complex shape of units together with their random placement makes calculation of the reaction forces between adjacent armor units impossible.

Consequently, deterministic calculations of the instantaneous armor unit stability conditions cannot be performed, which is why stability formulae are based on hydraulic model tests.

## Design of Rubble Mound Breakwaters

### Sea State Parameters

Wave height  
Wave length  
Wave steepness  
Shape of wave spectrum  
Water depth  
Wave incident angle  
Number of waves  
Mass density of water

### Structural Parameters

Slope of the structure  
Mass density of armour units  
Grading of rock armour,  $D_{n50}$ ,  $D_{n15}$ ,  $D_{n85}$   
Placement pattern, thickness of armour layer  
Porosity and permeability of armour, filter and core layers

## Design of Rubble Mound Breakwaters

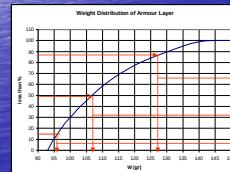
$D$  = diameter of the stone



How to determine diameter of the stone?



$D_n$  = nominal diameter  $= (W/\gamma_s)^{1/3}$  = equivalent cube length



## Design of Rubble Mound Breakwaters

$$W = \frac{\gamma_s H_D^3}{K_D \cot \alpha \left( \frac{\gamma_s}{\gamma_w} - 1 \right)^3}$$

Hudson formula

|            |                                                         |                     |
|------------|---------------------------------------------------------|---------------------|
| $W$        | = Weight of individual armour unit                      | (N)                 |
| $\gamma_s$ | = Unit weight of rock                                   | (N/m <sup>3</sup> ) |
| $H_{D=0}$  | = Design wave height for zero damage (0-5% damage)      | (m)                 |
| $K_D$      | = Damage coefficient or empirical stability coefficient |                     |
| $\gamma_w$ | = Unit weight of water                                  | (N/m <sup>3</sup> ) |
| $\alpha$   | = Angle between breakwater and horizontal               | (degrees)           |

## Design of Rubble Mound Breakwaters

- Hudson Formula can be written as following:

$$\frac{H_D}{\Delta D_n} = N_s = (K_D \cot \alpha)^{1/3}$$

where:

$\Delta$  = Relative mass density =  $(\gamma_s - \gamma_w) / \gamma_w$

$D_n$  = Nominal diameter of stone =  $(W/\gamma_s)^{1/3}$

## Kd values

| No-Damage Criteria and Riser Overlapping |     |           |                             |                   |                |                  |                  |
|------------------------------------------|-----|-----------|-----------------------------|-------------------|----------------|------------------|------------------|
| Armor Data                               | n   | Placement | Structure Trunk             |                   | Structure Head |                  | Slope<br>Oct     |
|                                          |     |           | K <sub>p</sub> <sup>2</sup> |                   | K <sub>p</sub> |                  |                  |
|                                          |     |           | Breaking Wave               | Nonbreaking Wave  | Breaking Wave  | Nonbreaking Wave |                  |
| Quarrystone                              | 2   | Random    | 3.3                         | 2.4               | 1.1            | 1.9              | 1.5 to 3.0       |
| Smooth rounded                           | >3  | Random    | 1.2                         | 2.2               | 1.1            | 2.2              | 5                |
| Rough angular                            | 1   | Random    | 2.0                         | 4.0               | 1.8            | 2.5              | 3.0              |
| Rough angular                            | 2   | Random    | 2.0                         | 4.0               | 1.8            | 2.5              | 3.0              |
| Rough angular                            | >3  | Random    | 2.2                         | 4.6               | 2.1            | 4.2              | 5                |
| Smooth angular                           | 2   | Special   | 3.8                         | 7.0               | 4.8            | 6.4              | 5                |
| Parallellepped                           | 2   | Special   | 7.0 - 20.0                  | 8.8 - 28.0        | ---            | ---              | ---              |
| Tetrapod                                 | 2   | Random    | 7.0                         | 8.0               | 6.0            | 6.0              | 1.5              |
| Quarried                                 | 2   | Random    | 8.0                         | 10.0              | 7.8            | 8.5              | 2.0              |
| Trihar                                   | 2   | Random    | 8.0                         | 10.0              | 7.8            | 8.5              | 2.0              |
| Dolce                                    | 2   | Random    | 15.8 <sup>8</sup>           | 31.8 <sup>8</sup> | 8.0            | 16.0             | 2.0 <sup>8</sup> |
| Modified cube                            | 2   | Random    | 8.8                         | 7.5               | 8.8            | 7.0              | 5                |
| Hexagon                                  | 2   | Random    | 8.0                         | 8.5               | 8.0            | 7.0              | 5                |
| Trihar                                   | 1   | Uniform   | 12.0                        | 15.0              | 7.8            | 8.5              | 5                |
| Quarrystone (K <sub>pa</sub> )           | --- | Random    | 2.2                         | 2.5               | ---            | ---              | ---              |
| Crested angular                          | --- | Random    | 2.2                         | 2.5               | ---            | ---              | ---              |

## Which wave height?

- Breaking waves
- H<sub>b</sub> (breaking)
- H<sub>b</sub> (broken)
- Non-Breaking waves
- H<sub>1/10</sub> at structure depth
- H<sub>1/10</sub> = 1.27 H<sub>1/3</sub>

## REGULAR WAVE CONCEPT

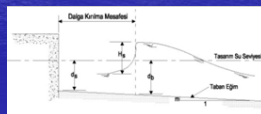
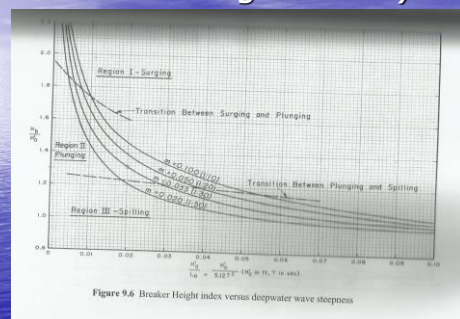
3 Cases for breaking

$ds > db$  nonbreaking  
 $d' < ds < db$  breaking  
 $ds < d'$  broken  
 $ds$  = structure construction depth

Breaker travel distance

$$X_p = (4.0 - 9.25m) H_b$$

$$d' = db - X_p \text{ m}$$

Is it a breaking wave? 1) Find H<sub>b</sub>



## example

- Find W of armour stone of trunk section by Hudson formula
- Given:  $H_0'=5\text{m}$ ,  $T=8\text{s}$ ,  $m=1/30$ ,  $d_s=6\text{m}$ ,  
 $\gamma_s=2.65\text{t/m}^3$ ,  $\gamma_w=1\text{t/m}^3$

Concrete armour

If it is tetrapod,  $K_D=7$

$$W = \frac{\gamma_s H_0'^3}{K_D \cot \alpha \left( \frac{\gamma_s}{\gamma_w} - 1 \right)^3} = \frac{2.3 \times 5.7^3}{7 \times 2 \times 1.3^3} = 13.8 \text{ t}$$